Predicting 30-day hospital readmission in patients with diabetes: a comparative machine learning study

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I. Introduction and Clinical Significance

Hospital readmission within 30 days of discharge is an important healthcare challenge, contributing to higher healthcare costs[1, 2]. Patients with diabetes are particularly vulnerable, with reported readmissions ranging between 14-20%, nearly twice those of patients without diabetes[1]. Factors associated with readmission include comorbidity burden, socioeconomic characteristics, and previous hospitalizations, highlighting the complexity of managing this population[1]. Additionally, transitioning from inpatient to outpatient care requires appropriate discharge planning, timely follow-up and medication management to reduce avoidable readmissions[1, 2].

Early identification of vulnerable patients could help healthcare professionals optimize resource allocation and targeted interventions. Recent studies have explored statistical and machine-learning approaches to predicting 30-day readmission among patients with diabetes, showing their potential and highlighting potential challenges in predictive performance[3, 4].

Therefore, this study aims to develop and compare statistical and machine-learning models for predicting 30-day hospital readmission among patients with diabetes using routinely collected clinical data, while emphasizing robust evaluation and model interpretability.

II. Literature Review and Related Work

Previous studies have identified several clinical and demographic factors associated with 30-day hospital readmission among patients with diabetes. These include age, sex, race, previous inpatient, outpatient and emergency department visits, and comorbidity burden, represented partly by recorded diagnoses[4-6]. Hospitalization characteristics, including admission type, lengths of stay, discharge plan and number of procedures and laboratory tests, have also been examined[5, 6]. Similarly, medication related variables, including insulin and oral antidiabetic therapy use, dosage changes, have been considered as indicators of diabetes management[5, 6].

Glycemic indicators have also been considered such as hemoglobin A1c (HbA1c), which reflects average blood glucose over approximately 3 months and serum glucose testing have also been investigated[5, 7]. Together, these findings support the selection of demographic, clinical, hospitalization, and medication-related variables in our prediction models.

Beyond Identifying relevant predictors, previous research has evaluated different approaches to predicting readmission. Tan et al. used logistic regression to predict prolonged hospitalization and 30-day readmission among elderly patients with T2DM. Their area under the receiver operating characteristic curve (AUROC) measures how well a model distinguishes between readmitted and non-readmitted patients. The AUROCmodel decreased from 0.769 internally to 0.633-0.671 in external population[3].

Studies using the University of California, Irvine (UCI) diabetes dataset have evaluated several approaches to predicting hospital readmission [4-6]. These include logistic regression, random forest, extreme gradient boosting (XGBoost), and a deep neural network [4-6]. Random forest combines multiple decision trees to make predictions, while XGBoost builds decision trees sequentially to improve prediction. Conversely, deep neural networks learn patterns through multiple layers of interconnected computational units[5, 6]. Emi-Johnson and Nkrumah reported AUROCs of 0.667 for XGBoost, 0.642 for logistic regression, and 0.579 for a deep neural network [4]. However, varied performances were reported across these approaches in other studies [8-10]. This suggests that a more complex model does not always improve predictions.

Differences in predictive performance across settings have also been shown in research involving pneumonia, colorectal surgery and other hospitalized populations [11-15]. This highlights the importance of testing models within specific clinical populations. In our dataset, approximately 11% of encounters resulted in readmission within 30 days, creating class imbalance, where non-readmission substantially outnumbers readmissions[7]. Therefore, overall accuracy may be misleading because the model could reach high accuracy by primarily predicting no readmission [16]. Another concern is data leakage, which may occur when encounters from the same patient are included in both model training and testing, which could possibly overestimate model performance[16]. To tackle these challenges, our project assigns all encounters from each patient to the same training, validation, or testing group. Model performance is evaluated using AUROC alongside the area

under the precision-recall curve (AUPRC), which describes the relationship between precision (the proportion of readmission predictions that were correct) and recall (the proportion of actual readmissions identified), making it particularly informative for imbalanced dataset[17].

Accordingly, our study aims to identify patients with diabetes at increased risk of 30-day readmission by comparing prediction models and examining the contributions of individual patient characteristics using SHapley Additive exPlanation (SHAP). The resulting model could ultimately support targeted discharge planning and timely follow-up [6, 18].

III. DESCRIPTION OF THE DATA SET

The project uses the Diabetes 130-US Hospital dataset (UCI Machine Learning Repository, dataset #296), covering inpatient encounters across 130 US hospitals between 1999 and 2008. The original dataset contains 101,766 encounters, 71,518 unique patients and 50 variables. After excluding hospice discharges and deaths (2,423 stays, 2.4%), the final analytical cohort consists of 69,990 patients, representing 99,343 encounters. The positive class, corresponding to the outcome of interest (30-day hospital readmission), represents 11.4% of this cohort.

Several variables contain missing data. Weight was missing in 96.9% of encounters, HbA1c test results were unavailable or not measured in 83.8% of encounters, and medical specialty was missing in 49.1%. A total of 23.5% of patients had more than one stay and accounted for 46% of rows (up to 40 stays per patient). Hence, rows are not independent, which motivates our patient-level split (see Evaluation Protocol) The population was 76.5% White, 19.3% Black, 2.0% Hispanic, 1.5% Other, and 0.6% Asian. Approximately 45% of the patients were 70 years old or over. The dataset relies on ICD-9 diagnosis codes, reflecting the coding practices used during the 1999-2008 study period. It does not include dates, hospital identifiers, vital signs, exact laboratory values, social factors or clinical notes. The positive rate differs little by sex (9.6%) or race (11.3%).

The dataset includes patient demographics, hospital admission and discharge characteristics, length of stay, and the number of laboratory tests, procedures, medications, and diagnoses recorded. It also contains healthcare utilization data from the preceding year (outpatient, emergency department, and inpatient visits), HbA1c and blood glucose test results, and 23 variables describing diabetes medications. Additionally, three ICD-9 diagnosis codes are classified into nine broader clinical categories, and three new variables are derived: total healthcare visits during the previous year, the number of medication changes, and an indicator combining HbA1c testing with changes in diabetes treatment.

Model training was feasible using standard CPU resources, without specialized hardware.

IV. EVALUATION PROTOCOL

The dataset was divided into training (70%, 69.519 encounters), validation (15%, 14.911 encounters), and testing (15%, 14.913 encounters) sets before model development. All encounters from the same patient were assigned to a single set to prevent data leakage. Model parameters were learned from the training set, while hyperparameters, classification thresholds, and calibration methods were selected using the validation set. The independent test set was reserved for a single final evaluation.

Given the class imbalance, model performance is primarily evaluated using the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC)[16, 17]. Accuracy, sensitivity and specificity are also reported at the selected classification threshold. The majority-class baseline achieves 88.9% accuracy but 0% sensitivity, highlighting the limitations of accuracy alone. To quantify uncertainty, 95% confidence intervals are estimated using patient-level bootstrap resampling, including paired comparisons between models.

Class weighting was applied to address class imbalance, while synthetic oversampling was also evaluated. Calibration is evaluated using reliability curves and the Brier score, while SHAP and decision-curve analysis assess model interpretability and potential clinical usefulness[18]. Performance is also evaluated across age and sex subgroups.

V. CANDIDATE MODELS

Several models of varying complexity were selected to predict 30-day hospital readmission among patients with diabetes. A majority-class baseline, which always predicts no readmission, serves as a reference for a comparison. Regularized logistic regression, a statistical approach that limits model complexity to reduce overfitting, was included as an interpretable model. Two gradient-boosting algorithms, extreme gradient boosting (XGBoost) and CatBoost, were selected for their ability to capture complex relationships between patient characteristics. CatBoost handles categorical variables directly, unlike XGBoost in our implementation, which requires one-hot encoding to convert categories into numerical indicators. A multilayer perceptron (MLP), a neural network with interconnected computational layers, was included to examine whether deep learning improves performance compared with tree-based methods[19]. Training, evaluation, calibration, and subgroup analyses have been implemented, with final results to be presented in the completed manuscript.

VI. WORK PLAN ROLES AND RISKS

Period	Planned work	Lead
Until Nov. 9	Complete the midterm report; ensure at least 15 references have been read by team members	Kayla and Magalie (Writing leads); all members
Nov. 10–24	Finalize model comparison; investigate sex-related performance differences; generate learning curves	Siham (Modelling lead)
Nov. 25–Dec. 3	Complete external-style validation checks; refine interface; prepare model card and required statements	Larissa (Interface lead); Siham (Data lead)

Dec. 4–10	Complete the 6–8-page manuscript; record the 7-minute presentation; prepare peer evaluations	All members
Dec. 11	Submit final deliverables; organize and finalize the repository	All members

Roles: Siham Khaldoun serves as both data and modeling lead, overseeing data preparation, model development, and evaluation. Kayla Arab and Magalie Achim jointly serve as writing leads, coordinating the literature review and manuscript preparation. Larissa Chiu serves as interface lead, overseeing interface development and potential deployment. All members contribute to both coding and scientific writing.

Risks: Limited clinical variables and the historical, US-specific dataset may restrict predictive performance and generalizability. Time constraints, data leakage, and class imbalance are addressed through coordinated responsibilities, patient-level separation, and appropriate evaluation metrics.

VII. CRediT Contribution Statement

Kayla Arab: Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Magalie Achim:** Investigation, Writing – original draft, Writing – review & editing. **Siham Khaldoun:** Data curation, Methodology, Software, Formal analysis, Validation, Writing – review & editing. **Larissa Chiu:** Investigation, Writing – review & editing.

VIII. Use of Generative AI Statement

The authors used ChatGPT (OpenAI, GPT-6) to assist with scientific writing, language editing, manuscript organization, and clarification of technical concepts. The tool was also used to identify potential scientific references and support the drafting of selected manuscript sections. Claude (Anthropic) was used to assist with dataset analysis and the selection of potential machine-learning models. All generated content was reviewed and revised by the authors. The authors take full responsibility for the accuracy and integrity of the submitted work.

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